

Term Project Report on

Daily Streamflow Prediction Using Traditional Machine Learning and Modern Time-Series Models with a Focus on Capturing High-Flow Events Relevant for Flood Prediction

by

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April 2026

Abstract

Accurate prediction of streamflow is essential for effective flood forecasting, water resource management, and hydropower planning, particularly in monsoon-dominated river basins. This study presents a comprehensive comparative analysis of machine learning (ML), deep learning (DL), and transformer-based models for daily streamflow prediction in the Mahanadi River basin.

Meteorological inputs, including rainfall and temperature obtained from the ERA5 dataset, along with observed discharge data from the Naraj station, were used for model development. Feature engineering techniques such as lagged discharge variables and cumulative rainfall indices (3-day and 7-day) were employed to capture hydrological processes and antecedent moisture conditions.

Four models - Random Forest, LightGBM, Long Short-Term Memory (LSTM), and a transformer-based model - were implemented and evaluated using Root Mean Square Error (RMSE), Nash-Sutcliffe Efficiency (NSE), and peak flow-specific metrics. The Random Forest model achieved the best overall performance (NSE = 0.878), outperforming both deep learning and transformer approaches. LSTM demonstrated reasonable performance (NSE $\approx$ 0.75) by capturing temporal patterns but showed limitations in predicting extreme peak flows. Transformer-based models exhibited comparatively lower performance due to data constraints and model complexity.

SHAP-based interpretability analysis revealed that short-term cumulative rainfall and antecedent discharge are the most influential predictors of streamflow. Additionally, peak flow evaluation indicated that while models effectively capture the timing of high-flow events, accurate prediction of peak magnitudes remains challenging.

The study highlights that, for moderate-sized hydrological datasets, feature engineering plays a more critical role than model complexity in improving predictive performance.

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Chapter - 1 : Introduction

1.1 Research Motivation

1.1.1 Mahanadi Basin Hydrology

Streamflow prediction is a key component of hydrological analysis, especially in river basins where rainfall patterns are highly variable and concentrated within short periods. In such systems, river discharge depends not only on current rainfall but also on past hydrological conditions such as antecedent rainfall and previous discharge.

The study focuses on the Mahanadi River Basin, which exhibits strong seasonal behavior driven by monsoon rainfall. A significant portion of annual discharge occurs within a few months, leading to sharp transitions between low-flow and high-flow conditions. This makes accurate prediction of streamflow, particularly during extreme events, both important and challenging.

The hydrological response of the basin is influenced by delayed runoff processes, accumulation effects, and persistence in flow. These characteristics make the system inherently nonlinear and time-dependent, requiring modeling approaches capable of capturing such dynamics.

1.1.2 Data-Driven Modeling Motivation

Traditional approaches to streamflow prediction often rely on physical or conceptual hydrological models. While these models provide interpretability, they require detailed information about basin characteristics and involve complex calibration procedures.

Data-driven approaches, particularly machine learning models, provide an alternative by learning relationships directly from data. These models are well-suited for capturing nonlinear interactions between variables such as rainfall and discharge without requiring explicit representation of physical processes.

Deep learning models extend this capability by learning temporal dependencies in sequential data. More recently, attention-based models have been explored for time-series prediction tasks. However, their effectiveness depends on data availability and the nature of the problem.

This study is motivated by the need to evaluate whether increasing model complexity actually improves prediction performance, especially when working with a single-station dataset.

1.2 Literature Review

Research on streamflow prediction has evolved from traditional statistical and physically-based models to data-driven approaches that leverage advances in machine

learning and deep learning. Early approaches such as regression models and time-series techniques were limited in their ability to capture nonlinear relationships and delayed hydrological responses, which are fundamental characteristics of river systems.

Machine learning models, particularly tree-based ensemble methods such as Random Forest and gradient boosting techniques, have been widely adopted due to their robustness and ability to model complex nonlinear interactions between input variables. These models perform well even with relatively small datasets and are less sensitive to noise and multicollinearity. A key advantage of such models is their compatibility with feature engineering, allowing incorporation of hydrological insights such as lagged discharge and accumulated rainfall.

Deep learning approaches, especially Long Short-Term Memory (LSTM) networks, have gained popularity for streamflow prediction due to their ability to model sequential dependencies. By maintaining memory of past inputs, LSTM models can capture temporal dynamics such as delayed runoff and persistence in flow. However, their performance is often dependent on the availability of large datasets and careful hyperparameter tuning. In cases with limited data, deep learning models may not consistently outperform simpler machine learning approaches.

More recently, transformer-based architectures have been introduced for time-series forecasting. These models rely on attention mechanisms to capture relationships across different time steps without relying solely on sequential processing. While transformers have shown promising results in domains such as natural language processing and large-scale forecasting, their application in hydrology is still emerging. Their effectiveness in streamflow prediction, particularly for single-station or moderate-sized datasets, remains uncertain.

Across the literature, model evaluation is typically based on aggregate performance metrics such as Nash-Sutcliffe Efficiency (NSE) and Root Mean Square Error (RMSE). While these metrics provide an overall measure of accuracy, they may not adequately reflect model performance during extreme events, which are of primary importance in flood forecasting. Additionally, many studies treat models as black boxes, with limited emphasis on interpretability and understanding the contribution of different input variables.

Another important observation is that while considerable effort has been directed toward developing more complex models, comparatively less attention has been given to the role of feature engineering. Hydrological processes are strongly influenced by antecedent conditions, and incorporating such information explicitly into the dataset can significantly improve model performance.

This study builds on these insights by not only comparing different modeling approaches but also emphasizing feature engineering, peak flow evaluation, and interpretability, thereby addressing several limitations observed in existing work.

1.2 Research Gaps

Based on the analysis of existing approaches, the following gaps are identified:

- Many studies prioritize model complexity over feature engineering
- Limited emphasis is placed on evaluating model performance during extreme flow events
- Comparative analysis across ML, DL, and transformer models is still limited in single-station setups
- Interpretability of model predictions is often not explored in detail

Additionally, there is a need to understand whether advanced models such as transformers provide meaningful improvements over simpler models when data availability is limited.

1.3 Objectives

The objectives of this study are:

- To develop and compare multiple modeling approaches including Random Forest, LightGBM, LSTM, and transformer models for streamflow prediction
- To evaluate model performance using metrics such as RMSE and NSE
- To analyze model performance specifically during peak flow conditions
- To examine the impact of feature engineering on predictive accuracy
- To interpret model behavior using SHAP-based analysis

1.4 Limitations

The study has the following limitations:

- The analysis is based on data from a single station, which may not fully capture spatial variability within the basin
- The dataset size is moderate, which may limit the performance of data-intensive models such as LSTM and transformers
- Extreme flow events are relatively rare, leading to imbalance in the dataset
- Only a limited set of input variables (rainfall, temperature, discharge) are considered, without explicit inclusion of physical basin characteristics

Chapter 2: Study Area and Data Collection

2.1 Mahanadi River Basin Overview

The study is conducted for the Mahanadi River Basin, one of the major river systems in eastern India. The basin originates in the highlands of Chhattisgarh and flows eastward through Odisha before draining into the Bay of Bengal.

The basin experiences a tropical monsoon climate, with a large proportion of annual rainfall concentrated between June and September. This seasonal concentration of rainfall leads to significant variability in river discharge, with relatively low flows during dry months and sharp increases during the monsoon season.

Hydrologically, the basin exhibits strong dependence on antecedent conditions. Runoff generation is influenced not only by current rainfall but also by cumulative rainfall over previous days and existing soil moisture conditions. This makes the system nonlinear and time-dependent, posing challenges for accurate streamflow prediction.

2.2 Naraj Station Overview

For this study, streamflow data from the Naraj station was used. The Naraj station is located near the downstream region of the basin and captures integrated discharge from a large upstream catchment.

Initially, a multi-station modeling approach was considered to better represent spatial variability across the basin. However, challenges related to data consistency, alignment across stations, and increased complexity led to the selection of a single representative station.

The Naraj station was chosen because it reflects basin-scale hydrological behavior and provides a consistent dataset suitable for modeling downstream flow conditions.

2.3 Data Sources

The dataset used in this study consists of both meteorological and hydrological variables.

Meteorological data, including rainfall and temperature, were obtained from the **ERA5 reanalysis dataset**. These data were originally available at high temporal resolution and were aggregated to daily values corresponding to the basin region.

Streamflow data obtained from **National Water Data Portal (NWDP)**, **National Water Informatics Centre (NWIC)** observed discharge records at the Naraj station. These records represent daily discharge and serve as the target variable for prediction.

During data preparation, a mismatch in dataset lengths was observed, with the meteorological dataset containing one additional observation compared to the streamflow data. This issue

was resolved by aligning both datasets based on common dates and removing the extra entry to ensure consistency.

The final dataset spans the period from 2010 to 2020 and includes:

- Daily rainfall (mm)
- Daily temperature (°C)
- Daily discharge (m³/s)

Chapter 3: Data Preprocessing and Feature Engineering

3.1 Data Preprocessing

The raw dataset obtained from meteorological and hydrological sources required several preprocessing steps before it could be used for modeling.

The meteorological variables, originally available at higher temporal resolution, were converted to daily values. Rainfall was aggregated using daily sums, while temperature was averaged to represent daily conditions.

One of the first challenges encountered was a mismatch in the number of observations between the meteorological dataset and the streamflow dataset. The weather dataset contained one additional row compared to the discharge data. This mismatch led to alignment issues when combining the datasets. To resolve this, both datasets were aligned based on common date indices, and the extra observation was removed to ensure consistency.

Another issue arose during exploratory data analysis, where unusually high rainfall values were observed. Initially, these values appeared inconsistent with expected ranges. However, after verifying the source and aggregation process, it was concluded that these values correspond to actual extreme rainfall events. Since such events are critical for modeling peak flows, they were retained in the dataset.

Temporal alignment between rainfall and discharge was carefully handled. Since streamflow response is influenced by prior rainfall, no artificial shifting was introduced during preprocessing. Instead, temporal dependencies were later incorporated through feature engineering.

The final preprocessed dataset consisted of aligned daily values of rainfall, temperature, and discharge, ensuring consistency across all variables.

3.2 Feature Construction

Feature engineering played a central role in this study, as it allowed incorporation of hydrological understanding into the modeling process.

To capture the persistence in streamflow, lagged discharge variables were created. These features represent the influence of previous discharge values on current flow conditions and help the model learn delayed system responses.

Rainfall was further processed to capture cumulative effects. Rolling sums of rainfall over multiple time windows, such as 3-day and 7-day periods, were computed. These features represent antecedent moisture conditions, which are critical in determining runoff generation.

Initially, additional features such as monsoon indicators were considered. However, these were not included in the final dataset, as seasonal patterns were already effectively captured through variables such as month and day of the year.

A binary feature representing heavy rainfall events was also included to capture extreme precipitation conditions.

Through this process, the dataset was transformed from simple raw inputs into a feature-rich representation of the hydrological system.

3.3 Model Development

3.3.1 Random Forest

Random Forest was used as the primary baseline model due to its robustness and ability to handle nonlinear relationships.

Random Forest is an ensemble learning method that constructs multiple decision trees and aggregates their predictions. Each tree is trained on a bootstrapped subset of the data, and at each split, a random subset of features is considered.

The final prediction is given by:

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N T_i(x)$$

where $T_i(x)$ represents the prediction from the i^{th} tree.

Implementation Details

- Input features: engineered dataset (rainfall, temperature, lags, rolling rainfall, etc.)
- Target variable: daily discharge
- No scaling required (tree-based model)

Key Parameters Used

- Number of trees: 200
- Maximum depth: 10
- random state: 42
- n_jobs: 1

3.3.2 LightGBM

LightGBM was implemented as a gradient boosting alternative to Random Forest.

LightGBM is a boosting algorithm that builds trees sequentially, where each new tree corrects the errors of previous trees.

The model minimizes a loss function using gradient descent:

$$L = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Each iteration updates predictions based on residual errors.

Implementation Details

- Same input features as Random Forest
- Faster training due to histogram-based splitting

Key Parameters Used

- n_estimators=1000,
- learning_rate=0.03,
- max_depth=5,
- num_leaves=25,
- subsample=0.8,
- colsample_bytree=0.8,
- random_state=42

3.3.3 LSTM

LSTM is a recurrent neural network designed to retain long-term dependencies using gated mechanisms.

The key equations include:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$

where f_t , i_t , and C_t represent forget, input, and cell states.

Data Preparation

Unlike tree-based models, LSTM requires sequential input.

A sliding window approach was used:

- Window size = 7 days
- Input: past 7 days
- Output: next day discharge

Model Architecture

- Input layer (sequence input)
- LSTM layer
- Dense output layer

Key Parameters Used

- Window size: 7
- Epochs: ~20
- Batch size: 32
- validation_split = 0.1
- Optimizer: Adam

3.3.4 Transformer Model

Transformers use attention mechanisms to weigh the importance of different time steps:

$$Attention(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

This allows the model to capture relationships across the entire sequence.

Due to limitations in implementing full Temporal Fusion Transformer (TFT), a simplified transformer encoder was used.

Key Parameters Used

- Number of attention heads: 4
- Hidden dimension: 32
- Dropout: 0.1

Challenges

- Library compatibility issues prevented TFT implementation
- Transformer required more data for stable learning

3.4 Evaluation Metrics

The performance of the developed models was evaluated using multiple metrics to assess both overall prediction accuracy and the ability to capture variability in streamflow.

3.4.1 Root Mean Square Error (RMSE)

Root Mean Square Error (RMSE) measures the average magnitude of prediction errors. It is defined as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Q_{obs,i} - Q_{pred,i})^2}$$

where $Q_{obs,i}$ and $Q_{pred,i}$ represent the observed and predicted discharge values, respectively, and n is the number of observations.

RMSE provides an absolute measure of error in the same units as the discharge (m^3/s). Lower RMSE values indicate better model performance. However, RMSE is sensitive to large errors and tends to emphasize deviations during extreme events.

3.4.2 Nash–Sutcliffe Efficiency (NSE)

Nash–Sutcliffe Efficiency (NSE) is a widely used metric in hydrology to evaluate model performance relative to the mean of observed data. It is defined as:

$$NSE = 1 - \frac{\sum_{i=1}^n (Q_{obs,i} - Q_{pred,i})^2}{\sum_{i=1}^n (Q_{obs,i} - \bar{Q}_{obs})^2}$$

where $\bar{Q}_{obs}$ is the mean observed discharge.

The NSE value ranges from $-\infty$ to 1:

- NSE=1: Perfect prediction
- NSE=0 : Model performs as well as mean of observations
- NSE<0 : Model performs worse than mean
- $0 < NSE < 1$: Generally acceptable levels of model performance, with higher values indicating better fit.

NSE is particularly useful in hydrology as it evaluates how well the model captures the variability of streamflow.

3.4.3 Peak Flow Evaluation Metrics

In addition to overall performance, model accuracy during high-flow events was evaluated separately.

Peak flow events were identified based on a threshold corresponding to high discharge values. For this subset of data, RMSE and NSE were computed separately to assess model performance during extreme conditions.

This approach allows evaluation of model capability in predicting flood-like conditions, which are of practical importance in hydrological applications.

Chapter 4: Results

4.1 Overall Model Performance

This section presents the performance of the implemented models for daily streamflow prediction. All models were trained and evaluated on the same dataset using consistent input features and evaluation metrics.

Model performance was assessed using Root Mean Square Error (RMSE) and Nash-Sutcliffe Efficiency (NSE). These metrics provide a measure of prediction accuracy and the ability of the model to capture variability in observed discharge.

The results indicate variation in performance across different modeling approaches, highlighting the influence of model structure and data representation.

Model	NSE	RMSE
Random Forest	0.878	1352
LightGBM	0.836	1570
LSTM	0.753	1931
Transformer	0.585	2500

4.2 Performance of Each Model

4.2.1 Random Forest

The Random Forest model achieved the best overall performance among all models.

- RMSE: 1352.49
- NSE: 0.878

The model demonstrated strong predictive capability, effectively capturing both general trends and variability in streamflow. The high NSE value indicates that the model closely follows observed discharge patterns.

4.2.2 LightGBM

The LightGBM model also showed strong performance, though slightly lower than Random Forest.

- RMSE: 1569.79
- NSE: 0.836

The model was effective in capturing overall trends but showed slightly higher error compared to Random Forest.

4.2.3 LSTM

The LSTM model was implemented to capture temporal dependencies in the data.

- RMSE: 1930.88
- NSE: 0.753

The model was able to capture general temporal patterns in streamflow but exhibited reduced accuracy compared to machine learning models. The performance gap suggests limitations in learning from the available dataset size.

4.2.4 Transformer Model

The transformer-based model showed the lowest performance among all approaches.

- RMSE: 2500.19
- NSE: 0.585

The model struggled to generalize effectively, likely due to higher model complexity and limited data availability. While it captured some temporal structure, prediction accuracy remained lower compared to other models.

4.3 SHAP-Based Interpretability Analysis

To understand the contribution of input features to model predictions, SHAP (SHapley Additive exPlanations) analysis was performed on the Random Forest model.

The SHAP summary results indicate that:

- Cumulative rainfall over short durations (3-day rainfall) is the most influential feature
- Lagged discharge variables significantly contribute to prediction accuracy
- Temperature has relatively lower influence compared to rainfall-related variables

The analysis confirms that both antecedent rainfall and previous discharge play a key role in determining current streamflow.

4.4 Peak Flow Analysis

In addition to overall performance, model accuracy during high-flow events was evaluated separately.

Peak flow events were identified based on high discharge values, and performance was measured using Peak NSE and Peak RMSE.

For the best-performing model:

- Peak NSE: 0.569
- Peak RMSE: 3826.92

The results indicate that while models are able to capture the occurrence and timing of high-flow events, there is a noticeable reduction in accuracy when predicting peak magnitudes.

This highlights the increased difficulty associated with modeling extreme hydrological events.

Chapter 5: Discussion

5.1 Superiority of Tree-Based Models for the Given Dataset

The results demonstrate that tree-based machine learning models, particularly Random Forest, achieved the highest predictive accuracy among all approaches. This indicates that, for the given dataset, models that rely on structured feature relationships outperform those that depend on learning complex representations.

The strong performance of Random Forest suggests that the relationships between rainfall, discharge, and engineered features are sufficiently captured through decision-tree splits, without requiring deeper temporal abstraction.

5.2 Feature Engineering as the Primary Driver of Model Performance

A key finding of this study is that feature engineering had a dominant influence on model performance.

The inclusion of lagged discharge variables and cumulative rainfall features significantly improved prediction accuracy. These features explicitly encode hydrological processes such as delayed runoff and antecedent moisture conditions.

The effectiveness of these features indicates that incorporating domain knowledge into the dataset reduces the burden on models to learn such relationships implicitly. This explains why even relatively simple models performed exceptionally well.

5.3 Limitations of Deep Learning under Moderate Data Conditions

The LSTM model, while capable of capturing temporal dependencies, did not outperform machine learning models in this study.

This can be attributed primarily to the size and nature of the dataset. Deep learning models typically require large volumes of data to generalize effectively. With a relatively limited dataset, the LSTM model was unable to fully leverage its representational capacity.

Additionally, the model exhibited smoothing behavior, particularly during high-flow events, indicating a tendency to prioritize overall error minimization over accurate prediction of extremes.

5.4 Ineffectiveness of Transformer-Based Models in This Setting

The transformer-based model showed the lowest performance among all approaches.

Although transformers are designed to capture long-range dependencies using attention mechanisms, their effectiveness depends heavily on large datasets and appropriate

architecture design. In this study, the available data was insufficient to support such complex modeling.

Furthermore, the simplified implementation used here may not fully represent the capabilities of advanced transformer architectures, but the results nonetheless indicate that increased model complexity does not guarantee improved performance.

5.5 Challenges in Modeling Peak Flow Magnitudes

A significant observation from the results is the reduced accuracy of all models during peak flow events.

While models were generally able to identify the occurrence and timing of high-flow events, they struggled to accurately estimate peak magnitudes. This is likely due to the rarity and variability of such events in the dataset.

Extreme flows are governed by nonlinear threshold processes, where small changes in rainfall can result in disproportionately large changes in discharge. Capturing such behavior is inherently difficult for data-driven models, especially when training data is limited.

5.6 Hydrological Consistency of Model Interpretability Results

The SHAP-based analysis provided insights that are consistent with known hydrological behavior.

Short-term cumulative rainfall (3-day rainfall) emerged as the most influential predictor, followed by lagged discharge variables. These findings align with the understanding that runoff generation depends on both recent rainfall and existing flow conditions.

The relatively lower importance of temperature suggests that, in this basin, streamflow variability is primarily driven by rainfall dynamics.

5.7 Influence of Dataset Characteristics on Model Behavior

The characteristics of the dataset played a crucial role in determining model performance.

The moderate size of the dataset favored models that can perform well with limited data, such as Random Forest. In contrast, models that require extensive training data, such as LSTM and transformers, were constrained.

Additionally, the imbalance between normal and extreme flow conditions influenced model predictions, particularly in peak flow estimation.

5.8 Implications for Model Selection in Hydrological Applications

The findings of this study suggest that model selection should be guided by data characteristics and problem requirements rather than model complexity.

For moderate-sized datasets, tree-based models combined with well-designed features provide a reliable and efficient solution. While deep learning and transformer models offer theoretical advantages, their practical benefits may not be realized without sufficient data.

These insights are particularly relevant for real-world hydrological applications, where data availability is often limited.

Chapter 6: Conclusion

6.1 Conclusions

This study presented a comparative analysis of machine learning, deep learning, and transformer-based models for daily streamflow prediction in the Mahanadi River Basin using meteorological and hydrological data.

The results demonstrated that tree-based machine learning models, particularly Random Forest, achieved the highest predictive performance, with an NSE of 0.878. LightGBM also showed strong performance, while LSTM and transformer-based models were comparatively less effective under the given data conditions.

A key finding of this study is that feature engineering played a more significant role than model complexity. The inclusion of lagged discharge and cumulative rainfall features enabled models to effectively capture hydrological dynamics such as delayed runoff and antecedent moisture conditions.

The analysis further revealed that while models were able to capture overall trends and the timing of flow variations, accurate prediction of peak flow magnitudes remains a challenge. This limitation highlights the difficulty of modeling extreme hydrological events using data-driven approaches.

Interpretability analysis using SHAP confirmed that short-term cumulative rainfall and previous discharge are the most influential factors governing streamflow. These findings are consistent with established hydrological understanding and validate the modeling approach adopted in this study.

Overall, the study demonstrates that for moderate-sized datasets, simpler models combined with domain-informed features provide reliable and efficient solutions for streamflow prediction.

6.2 Limitations

Despite the insights obtained, the study has certain limitations.

The analysis is based on data from a single station, which may not fully capture spatial variability across the basin. Additionally, the dataset size is relatively limited, which affects the performance of data-intensive models such as LSTM and transformers.

Extreme flow events are relatively infrequent, leading to imbalance in the dataset and reduced accuracy during peak flow prediction. Furthermore, the study relies on a limited set of input variables and does not explicitly incorporate physical basin characteristics such as land use or soil properties.

6.3 Future Work

Future research can extend this work in several directions.

- Expanding the analysis to multiple stations within the basin to capture spatial variability
- Incorporating additional variables such as soil moisture, land use, and evapotranspiration
- Exploring advanced architectures such as Temporal Fusion Transformer (TFT) with larger datasets
- Developing hybrid models that combine physical understanding with data-driven approaches
- Improving peak flow prediction through specialized modeling techniques or data augmentation

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Appendix A

Supplementary Figures

	date	rainfall_mm	temperature_C	discharge
count	4017	4017.000000	4017.000000	4017.000000
mean	2015-07-02 00:00:00.000000256	4.331058	26.054813	2059.732532
min	2010-01-01 00:00:00	0.000000	14.914246	0.000000
25%	2012-10-01 00:00:00	0.021707	23.201782	177.670000
50%	2015-07-02 00:00:00	0.690111	26.669708	333.490000
75%	2018-04-01 00:00:00	5.998534	28.472656	1569.320000
max	2020-12-30 00:00:00	60.209839	35.738953	161104.940000
std	NaN	7.188893	3.983835	5813.783625

Figure A.1 : Final dataset description

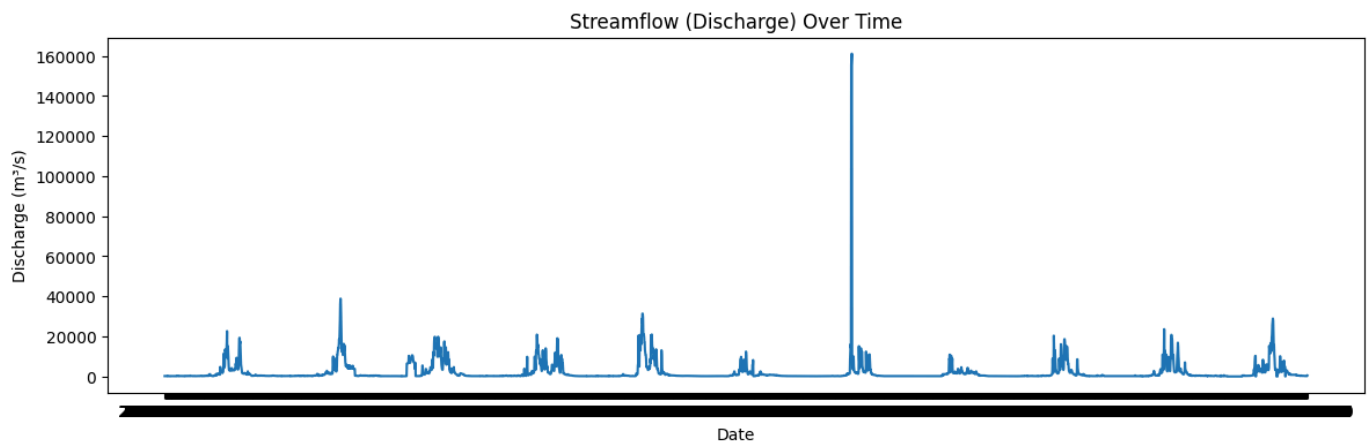


Figure A.2 : Plot of Discharge vs Time from year 2010 - 2020

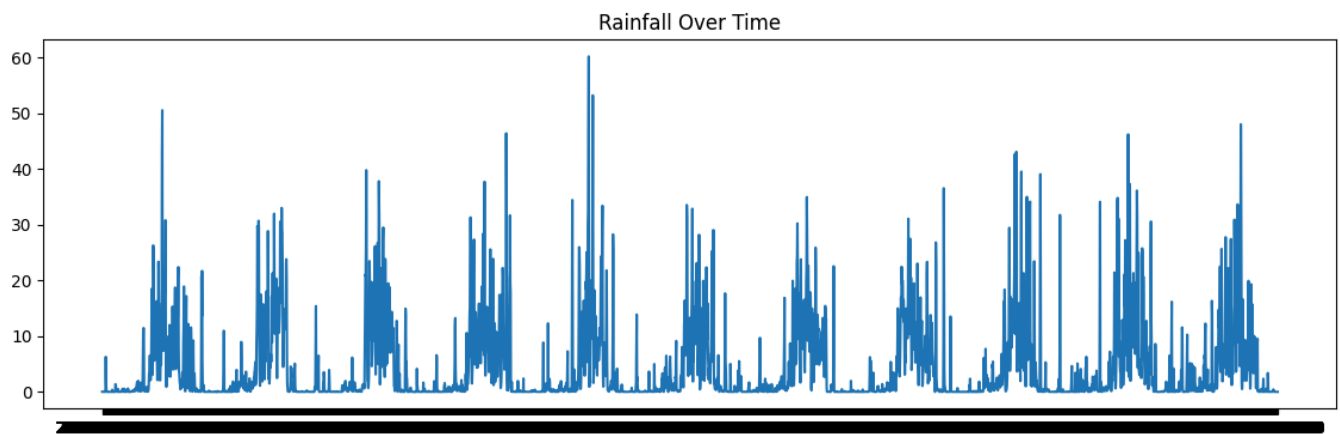


Figure A.3 : Plot of Rainfall vs Time from year 2010 - 2020

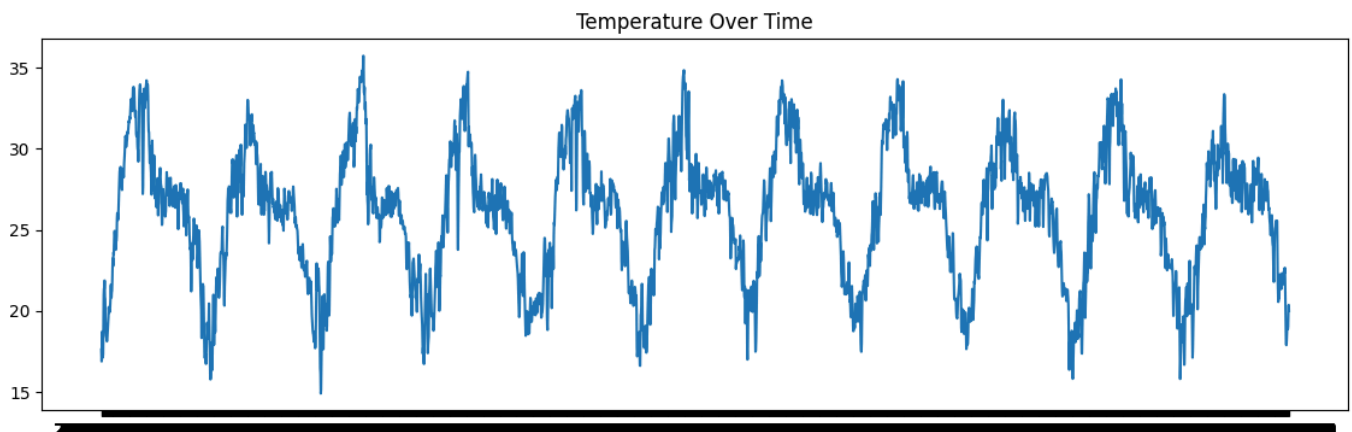


Figure A.4 : Plot of Temperature vs Time from year 2010 - 2020

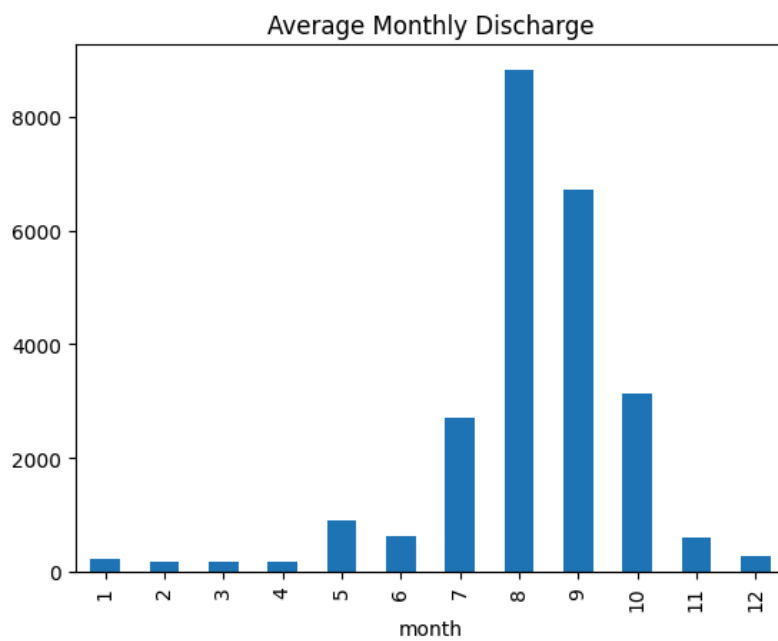


Figure A.5 :Average Monthly Discharge Plot

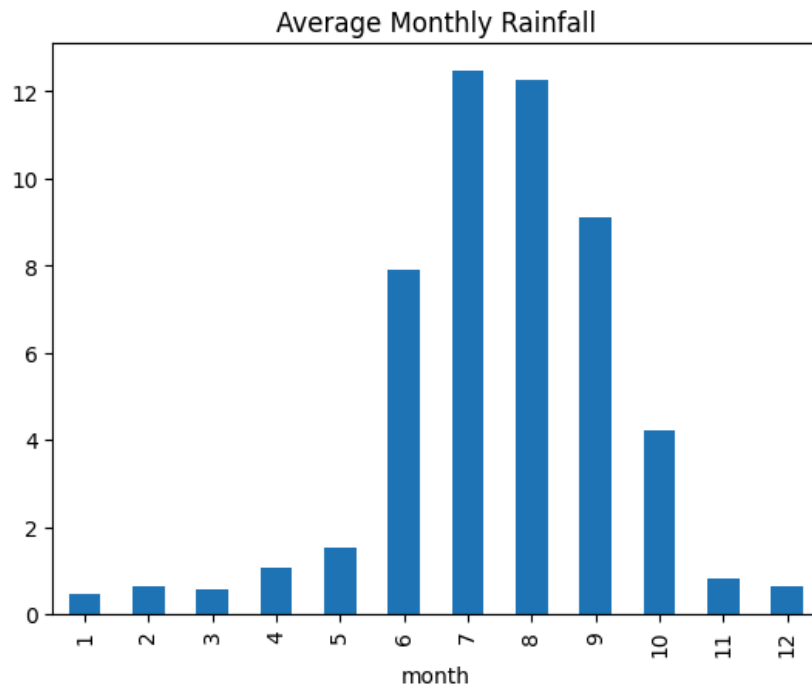


Figure A.6 :Average Monthly Rainfall Plot

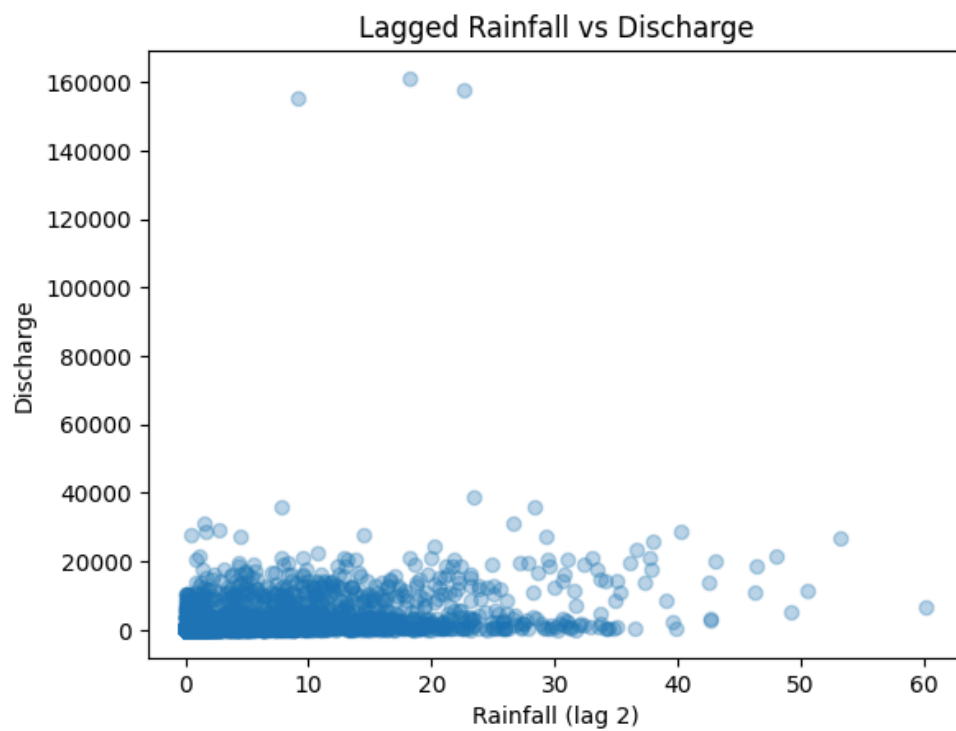


Figure A.7 : Lagged Rainfall vs Discharge Plot

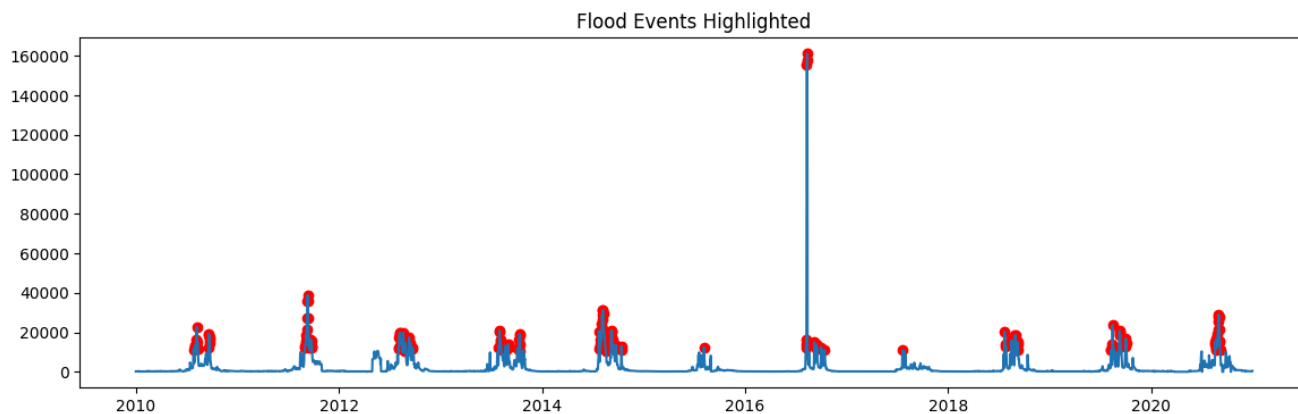


Figure A.8 : Peak Events Highlighted

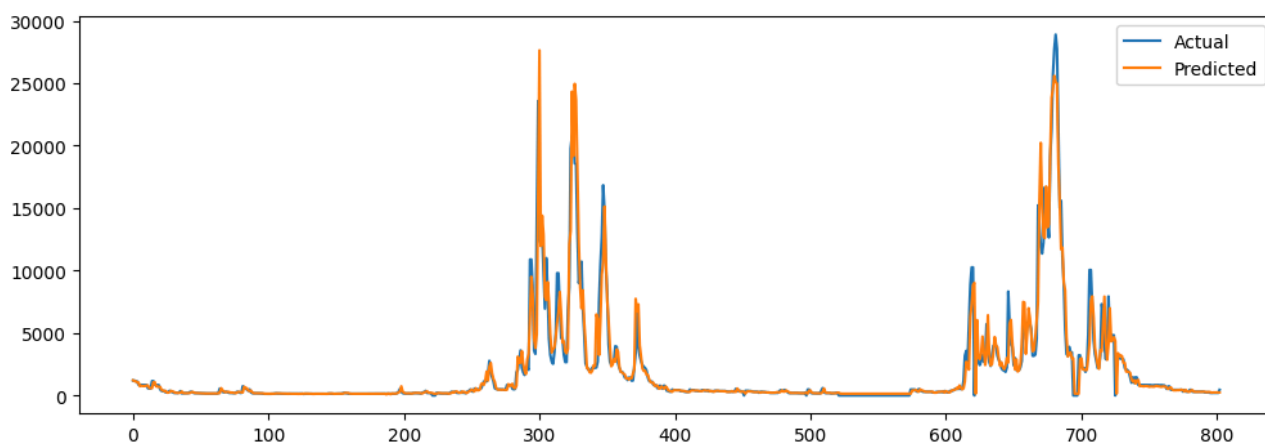


Figure A.9 : Random Forest Model Predictions

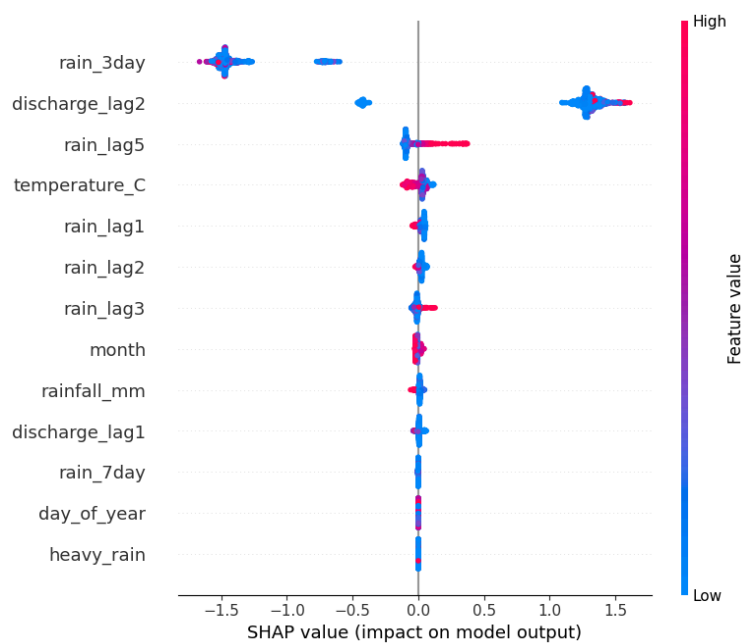


Figure A.10 : SHAP summary Plot on Random Forest Model

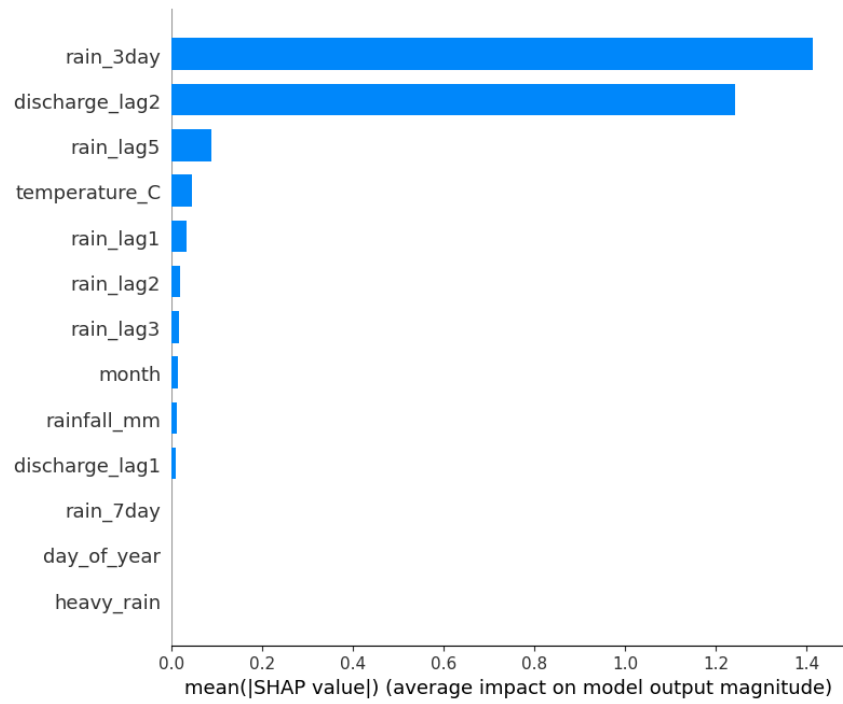


Figure A.11 : SHAP summary Plot on Random Forest Model

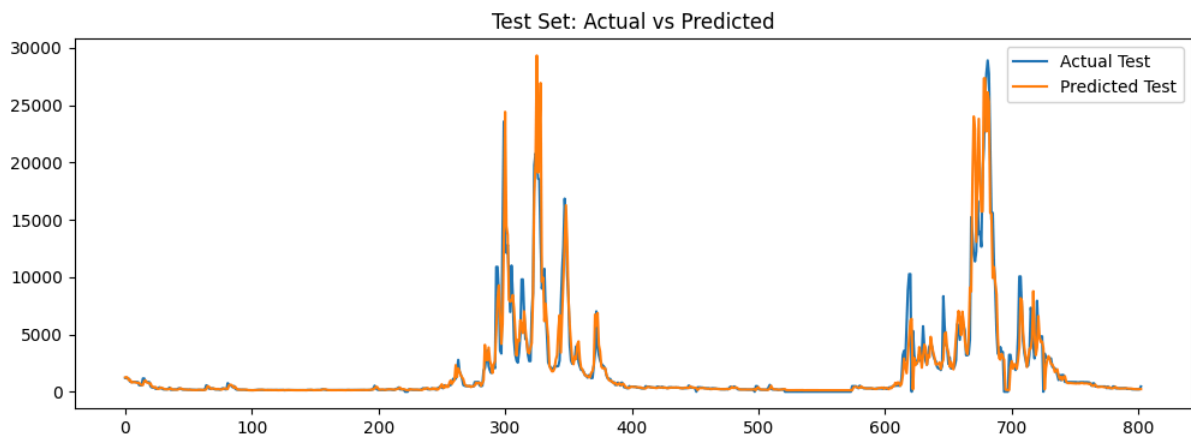


Figure A.12 : LightGBM Model Predictions

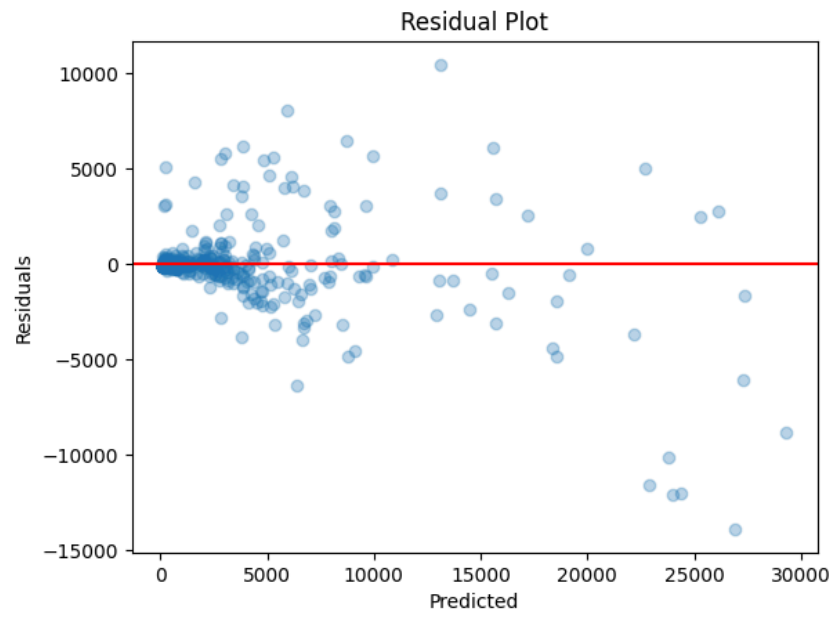


Figure A.13 : LightGBM Model Residual Plot

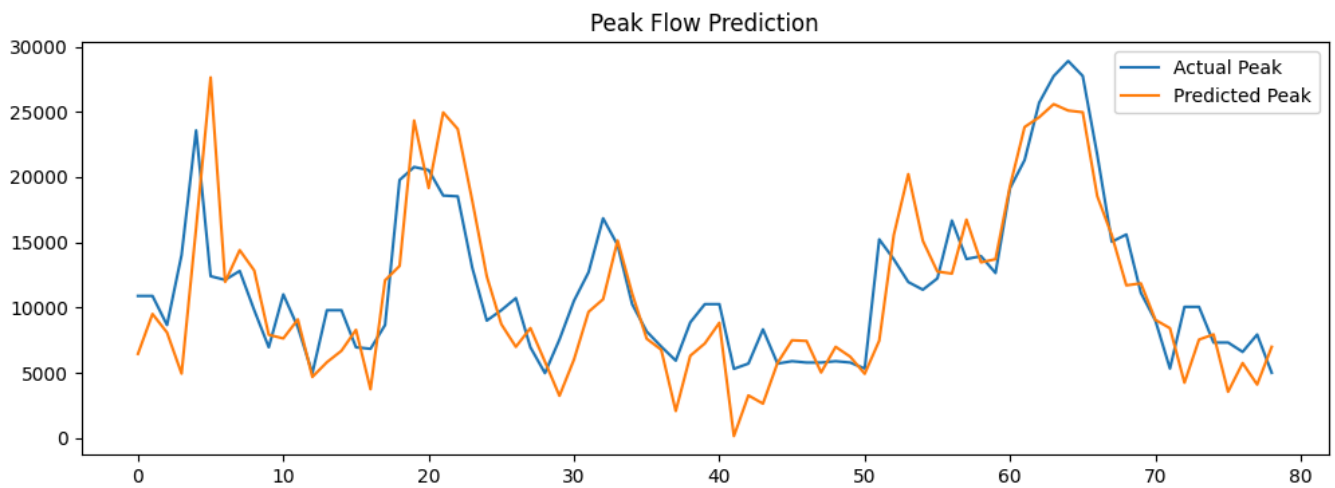


Figure A.14 : Peak Flow Prediction Plot

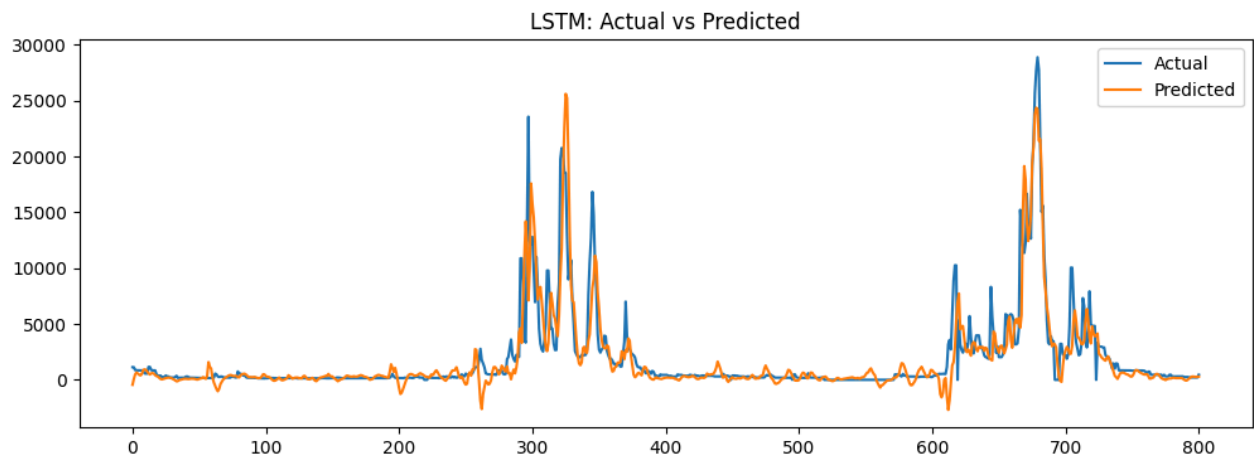


Figure A.15 : LSTM Model Prediction